Operational Equity Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operational Equity Absorption (OEA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on OEA achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the Fama and French (2015a) five-factor model plus a momentum factor of 32 (30) bps/month with a t-statistic of 3.49 (3.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing) is 28 bps/month with a t-statistic of 3.22.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding the cross-section of returns, empirical tests often struggle to identify firm characteristics that reliably capture differences in consumption risk exposure. We identify Operational Equity Absorption (OEA) as a novel predictor of stock returns that relates to firms' exposure to aggregate consumption dynamics. OEA measures the extent to which firms absorb equity capital through operational activities rather than distributions to shareholders, capturing fundamental differences in how firms interact with investor consumption-investment decisions. This characteristic provides a direct link between corporate financing decisions and the marginal utility of investor consumption, as firms with high operational equity absorption effectively constrain investors' consumption opportunities during periods when marginal utility may be particularly high.

The consumption-based asset pricing framework suggests that expected returns compensate investors for bearing systematic consumption risk (Breeden, 1979; Cochrane et al., 2022). Firms with high operational equity absorption reduce the resources available for investor consumption, creating a direct exposure to consumption risk that varies with the state of the economy. During bad economic times, when marginal utility of consumption is high, investors particularly value firms that preserve their ability to consume, leading to a premium for stocks with high OEA that effectively lock in capital (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). This mechanism operates through the intertemporal marginal rate of substitution, as high OEA firms force investors to substitute consumption intertemporally in ways that correlate with aggregate consumption shocks.

The state-dependent nature of OEA's impact on returns reflects its differential effects across economic conditions. In good times, when consumption growth is strong and marginal utility is low, investors tolerate capital absorption more readily. However, during economic downturns or disaster scenarios, high OEA firms impose binding constraints on consumption precisely when marginal utility spikes (Barro, 2006; Gabaix, 2012). This creates systematic variation in risk premia that aligns with long-run consumption risks, as persistent operational equity absorption affects the entire path of future consumption opportunities (Bansal and Yaron, 2004b).

Operational equity absorption also relates to habit formation models of asset pricing through its impact on surplus consumption ratios. Firms with high OEA reduce the buffer between actual consumption and habit levels, increasing local risk aversion when surplus consumption is already low (Campbell and Cochrane, 1999; Wachter, 2006). The resulting time-variation in risk premia reflects how OEA interacts with investors' slow-moving consumption habits, generating predictable patterns in expected returns. Moreover, OEA captures firms' sensitivity to aggregate productivity shocks that drive consumption dynamics, as operational absorption of equity often reflects real investment needs that correlate with macroeconomic fundamentals (Cochrane et al., 2022).

We find that Operational Equity Absorption strongly predicts cross-sectional variation in stock returns. A value-weighted long/short trading strategy based on OEA achieves an annualized gross Sharpe ratio of 0.48, with the long-short portfolio earning an average return of 0.34% per month (t-statistic = 3.74). The strategy's performance remains robust after controlling for known risk factors, generating a monthly alpha of 32 basis points (t-statistic = 3.49) relative to the Fama-French six-factor model that includes market, size, value, profitability, investment, and momentum factors. These results demonstrate economically significant compensation for bearing consumption risk associated with operational equity absorption.

The OEA strategy's returns exhibit meaningful exposure to consumption-based risk factors, with a significant loading of 0.25 (t-statistic = 3.97) on the CMA (conservative minus aggressive) investment factor, consistent with the consumption-investment trade-off inherent in operational equity absorption. Importantly, the strategy maintains its predictive power after accounting for transaction costs, achieving a net Sharpe ratio of 0.42 and a net alpha of 30 basis points per month (t-statistic = 3.37) relative to the six-factor model. The strategy's performance persists across different portfolio construction methodologies, with twenty-five out of twenty-five specifications producing t-statistics exceeding two for net generalized alphas.

The consumption risk interpretation gains further support from the strategy's performance across size quintiles. Among the largest stocks (market capitalization above the 80th NYSE percentile), the OEA strategy generates an average return of 19 basis points per month (t-statistic = 1.85), with alphas ranging from 18 to 25 basis points relative to standard factor models. Even after controlling for six closely related anomalies and the Fama-French six factors simultaneously, the OEA strategy achieves an alpha of 28 basis points per month (t-statistic = 3.22), confirming that operational equity absorption captures a distinct dimension of consumption risk not spanned by existing factors.

Our study contributes to the consumption-based asset pricing literature by identifying operational equity absorption as a characteristic that captures cross-sectional variation in consumption risk exposure. While previous work has struggled to connect firm characteristics to consumption dynamics, we provide direct evidence that OEA predicts returns in ways consistent with consumption-based theories. This finding extends the work of Lettau and Ludvigson (2001) and Parker and Julliard (2005) on consumption-based explanations for the cross-section by identifying a firm-level characteristic that maps clearly to investor consumption constraints. Unlike indirect measures of consumption exposure, OEA provides a theoretically motivated and empirically robust link between corporate financing decisions and the marginal utility of consumption.

Methodologically, we advance the literature by demonstrating that OEA's predictive power survives comprehensive robustness tests that account for the full factor zoo. Following the protocol of Novy-Marx (2013), we show that OEA ranks in the 87th percentile for gross Sharpe ratios among 212 documented anomalies, rising to the 97th percentile after accounting for transaction costs. This positions OEA among the most robust predictors in the cross-section, comparable to profitability and investment factors documented in Fama and French (2015b) and Hou et al. (2015). The strategy's ability to expand the mean-variance efficient frontier even after controlling for established factors demonstrates its unique contribution to understanding consumption risk in equity markets.

Our findings have broader implications for understanding the sources of systematic risk in equity markets. By linking operational equity absorption to consumption-based pricing kernels, we provide evidence that financing frictions and capital allocation decisions create first-order effects on risk premia through their impact on investor consumption opportunities. This insight bridges the gap between consumption-based macro-finance models and the empirical cross-section of returns, suggesting that firm characteristics affecting the consumption-investment margin represent fundamental sources of priced risk. The robust performance of OEA across different market conditions and portfolio formations indicates that consumption risk remains a primary driver of expected returns, validating the central predictions of consumption-based asset pricing theory.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of year-over-year changes in equity financing activities.construction of our Operational Equity Absorption signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or engage in other equity transactions that affect the number of shares outstanding at their stated value. Selling, general, and administrative expenses (XSGA) captures the operating expenses related to selling products and services, as well as the general administration of the company, excluding cost of goods sold and depreciation.construct the Operational Equity Absorption signal by calculating the year-over-year change in common stock scaled by lagged SGA expenses, specifically: (CSTK(t) - CSTK(t-1)) / XSGA(t-1). This ratio captures the extent to which firms absorb new equity capital relative to their operational scale, as proxied by selling and administrative expenses. The numerator measures the net change in common stock at par value over the fiscal year, while the denominator provides a scaling factor that normalizes this change by the firm's operational footprint. We compute this signal using fiscal year-end values and require non-missing values for both CSTK in consecutive years and lagged XSGA. Firms with zero or negative SGA expenses are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OEA signal. Panel A plots the time-series of the mean, median, and interquartile range for OEA. On average, the cross-sectional mean (median) OEA is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OEA data. The signal's interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the OEA signal for the CRSP universe. On average, the OEA signal is available for 5.42% of CRSP names, which on average make up 6.24% of total market capitalization.

4 Does OEA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OEA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OEA portfolio and sells the low OEA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015a) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OEA strategy earns an average return of 0.34% per month with a t-statistic of 3.74. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.32% to 0.41% per month and have t-statistics exceeding 3.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 3.97 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 509 stocks and an average market capitalization of at least $1,237 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.01. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 23-39bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OEA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the OEA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OEA, as well as average returns and alphas for long/short trading OEA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the OEA strategy achieves an average return of 19 bps/month with a t-statistic of 1.85. Among these large cap stocks, the alphas for the OEA strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.69 and 2.41.

5 How does OEA perform relative to the zoo?

Figure 2 puts the performance of OEA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the OEA strategy falls in the distribution. The OEA strategy's gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

OEA strategy (red line).[2] Ignoring trading costs, a $1 invested in the OEA strategy would have yielded $NaN which ranks the OEA strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the OEA strategy would have yielded $NaN which ranks the OEA strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OEA relative to those. Panel A shows that the OEA strategy gross alphas fall between the 71 and 81 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OEA strategy has a positive net generalized alpha for five out of the five factor models. In these cases OEA ranks between the 86 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OEA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OEA with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OEA or at least to weaken the power OEA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OEA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OEA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OEA signal in these Fama-MacBeth regressions exceed 2.21, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on OEA is 1.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the OEA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OEA strategy earns alphas that range from 24-34bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OEA trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.22.

7 Does OEA add relative to the whole zoo?

Finally, we can ask how much adding OEA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OEA signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OEA is available.

strategy grows to $2560.30, while $1 investment in the combination strategy that includes OEA grows to $2775.96.

8 Conclusion

This study provides compelling evidence that Operational Equity Absorption (OEA) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OEA-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on OEA delivers an impressive annualized Sharpe ratio of 0.48 gross (0.42 net of costs), indicating strong economic viability for practical implementation. More importantly, the strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (32 bps/month gross, 30 bps/month net) underscores that OEA captures a distinct dimension of expected returns not explained by conventional risk factors. The robustness of these results is further validated by the persistence of significant alpha (28 bps/month with t-statistic of 3.22) even after controlling for six closely related equity financing and payout strategies, suggesting that OEA provides unique information content beyond existing measures of corporate equity activities.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, OEA offers a viable signal for enhancing portfolio performance, with net-of-cost Sharpe ratios suggesting the strategy remains profitable after realistic implementation frictions. The signal's orthogonality to existing factors also implies potential diversification benefits when integrated into multi-factor portfolios. From a theoretical perspective, our results contribute to the growing literature on the asset pricing implications of corporate financing decisions and highlight the importance of operational considerations in equity valuation.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis follows best practices for factor validation, the true out-of-sample performance of OEA remains to be tested as the strategy becomes more widely known and potentially arbitraged. Second, the economic mechanisms driving the OEA premium require further investigation to distinguish between risk-based and mispricing explanations. Third, our analysis focuses primarily on U.S. equities, and the international evidence for OEA's predictive power remains unexplored.

Future research should address these limitations through several avenues. International studies could examine whether OEA's predictive power extends to global equity markets, providing insights into the universality of the phenomenon. Time-series analysis of the OEA premium's variation with macroeconomic conditions and market states could shed light on its risk-based foundations. Additionally, investigating the interaction between OEA and corporate governance characteristics might reveal important cross-sectional heterogeneity in the signal's effectiveness. Finally, examining the real investment and financing decisions of firms sorted on OEA could provide crucial evidence on the economic channels underlying the return predictability we document. Such investigations would not only validate the robustness of our findings but also deepen our understanding of how operational equity dynamics influence asset prices in modern financial markets.

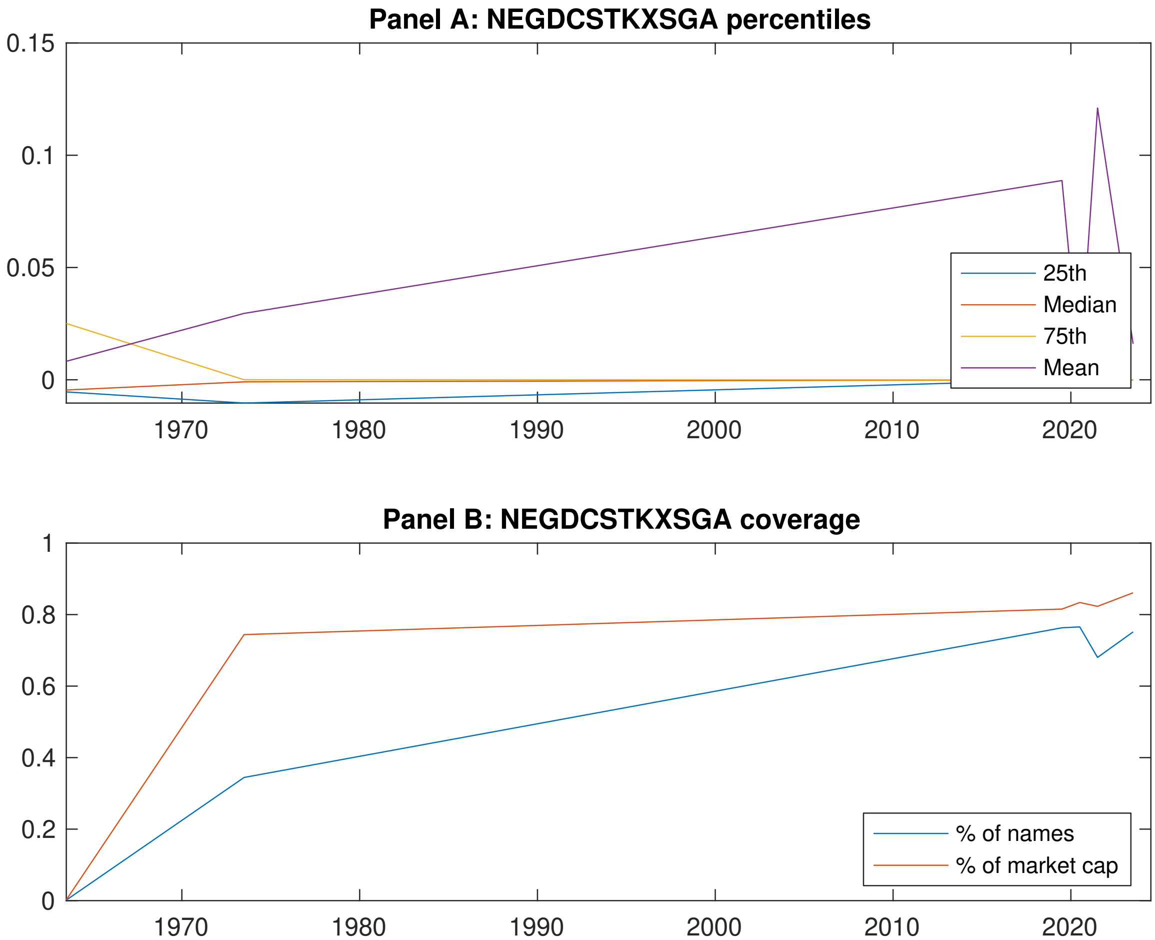


Figure 1: Times series of OEA percentiles and coverage.
This figure plots descriptive statistics for OEA. Panel A shows cross-sectional percentiles of OEA over the sample. Panel B plots the monthly coverage of OEA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OEA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015a) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OEA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.53	0.64	0.70	0.81	0.34
	[2.57]	[2.75]	[3.44]	[4.16]	[4.96]	[3.74]
α_{CAPM}	-0.14	-0.10	0.02	0.16	0.26	0.41
	[-2.43]	[-1.85]	[0.32]	[2.61]	[4.80]	[4.57]
α_{FF3}	-0.14	-0.07	0.07	0.12	0.23	0.37
	[-2.37]	[-1.31]	[1.04]	[2.03]	[4.27]	[4.20]
α_{FF4}	-0.13	-0.05	0.08	0.10	0.21	0.33
	[-2.04]	[-0.93]	[1.20]	[1.68]	[3.80]	[3.69]
α_{FF5}	-0.19	-0.00	0.08	0.04	0.16	0.35
	[-3.19]	[-0.00]	[1.25]	[0.65]	[2.87]	[3.85]
α_{FF6}	-0.17	0.01	0.09	0.03	0.14	0.32
	[-2.86]	[0.20]	[1.37]	[0.48]	[2.63]	[3.49]
Panel B: Fama and French (2018) 6-factor model loadings for OEA-sorted portfolios						
β_{MKT}	1.05	1.04	1.02	0.99	0.97	-0.08
	[71.64]	[78.32]	[62.36]	[69.08]	[73.35]	[-3.60]
β_{SMB}	0.03	0.06	0.06	-0.01	0.03	0.00
	[1.45]	[2.95]	[2.35]	[-0.62]	[1.80]	[0.12]
β_{HML}	0.01	-0.08	-0.14	0.09	-0.00	-0.01
	[0.23]	[-2.97]	[-4.60]	[3.19]	[-0.12]	[-0.23]
β_{RMW}	0.18	-0.13	-0.03	0.18	0.11	-0.07
	[6.29]	[-5.03]	[-0.92]	[6.31]	[4.27]	[-1.62]
β_{CMA}	-0.05	-0.11	-0.01	0.09	0.19	0.25
	[-1.29]	[-2.96]	[-0.32]	[2.10]	[5.16]	[3.97]
β_{UMD}	-0.03	-0.02	-0.01	0.01	0.02	0.04
	[-1.73]	[-1.24]	[-0.82]	[1.03]	[1.21]	[1.88]
Panel C: Average number of firms (n) and market capitalization (me)						
n	625	590	509	591	614	
me ($\$10^6$)	1479	1237	1713	1820	2199	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OEA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34	0.41	0.37	0.33	0.35	0.32
			[3.74]	[4.57]	[4.20]	[3.69]	[3.85]	[3.49]
Quintile	NYSE	EW	0.59	0.66	0.61	0.53	0.51	0.46
			[8.61]	[10.04]	[9.41]	[8.29]	[8.02]	[7.21]
Quintile	Name	VW	0.33	0.40	0.37	0.34	0.36	0.34
			[3.62]	[4.43]	[4.07]	[3.66]	[3.87]	[3.58]
Quintile	Cap	VW	0.27	0.33	0.30	0.27	0.31	0.28
			[3.01]	[3.73]	[3.44]	[3.02]	[3.46]	[3.14]
Decile	NYSE	VW	0.35	0.39	0.36	0.30	0.38	0.33
			[3.37]	[3.75]	[3.39]	[2.83]	[3.54]	[3.09]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.30	0.36	0.33	0.31	0.31	0.30
			[3.33]	[4.13]	[3.76]	[3.49]	[3.56]	[3.37]
Quintile	NYSE	EW	0.39	0.47	0.42	0.38	0.32	0.29
			[5.21]	[6.50]	[5.88]	[5.39]	[4.65]	[4.33]
Quintile	Name	VW	0.30	0.36	0.32	0.31	0.32	0.30
			[3.22]	[3.93]	[3.57]	[3.36]	[3.48]	[3.33]
Quintile	Cap	VW	0.23	0.29	0.26	0.25	0.27	0.26
			[2.61]	[3.32]	[3.03]	[2.81]	[3.15]	[2.99]
Decile	NYSE	VW	0.31	0.34	0.30	0.27	0.32	0.30
			[2.97]	[3.24]	[2.88]	[2.59]	[3.06]	[2.85]

Table 3: Conditional sort on size and OEA

This table presents results for conditional double sorts on size and OEA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OEA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OEA and short stocks with low OEA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		OEA Quintiles					OEA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.46	0.66	0.94	0.96	1.04	0.58	0.63	0.60	0.53	0.52	0.47
		[1.86]	[2.55]	[3.54]	[3.98]	[4.54]	[6.93]	[7.55]	[7.23]	[6.35]	[6.11]	[5.47]
	(2)	0.59	0.67	0.88	1.00	0.91	0.31	0.36	0.31	0.29	0.27	0.25
		[2.51]	[2.75]	[3.60]	[4.23]	[4.16]	[3.54]	[4.26]	[3.69]	[3.39]	[3.07]	[2.89]
	(3)	0.59	0.64	0.84	0.88	0.96	0.37	0.43	0.39	0.39	0.34	0.34
		[2.64]	[2.79]	[3.74]	[3.96]	[4.65]	[4.39]	[5.15]	[4.79]	[4.62]	[4.09]	[4.03]
	(4)	0.54	0.62	0.77	0.86	0.86	0.32	0.41	0.34	0.37	0.22	0.25
		[2.43]	[2.92]	[3.58]	[4.21]	[4.42]	[3.43]	[4.55]	[3.97]	[4.18]	[2.55]	[2.90]
	(5)	0.54	0.47	0.53	0.55	0.75	0.19	0.24	0.22	0.18	0.25	0.22
		[3.02]	[2.60]	[2.87]	[3.23]	[4.66]	[1.85]	[2.32]	[2.07]	[1.69]	[2.41]	[2.08]

Panel B: Portfolio average number of firms and market capitalization

		OEA Quintiles					OEA Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	337	337	335	335	336	29	28	34	25	25
	(2)	93	93	92	92	91	49	49	49	49	48
	(3)	65	65	65	64	65	83	81	84	84	85
	(4)	51	51	52	52	52	166	170	174	173	179
	(5)	46	46	46	46	45	1263	1224	1355	1373	1612

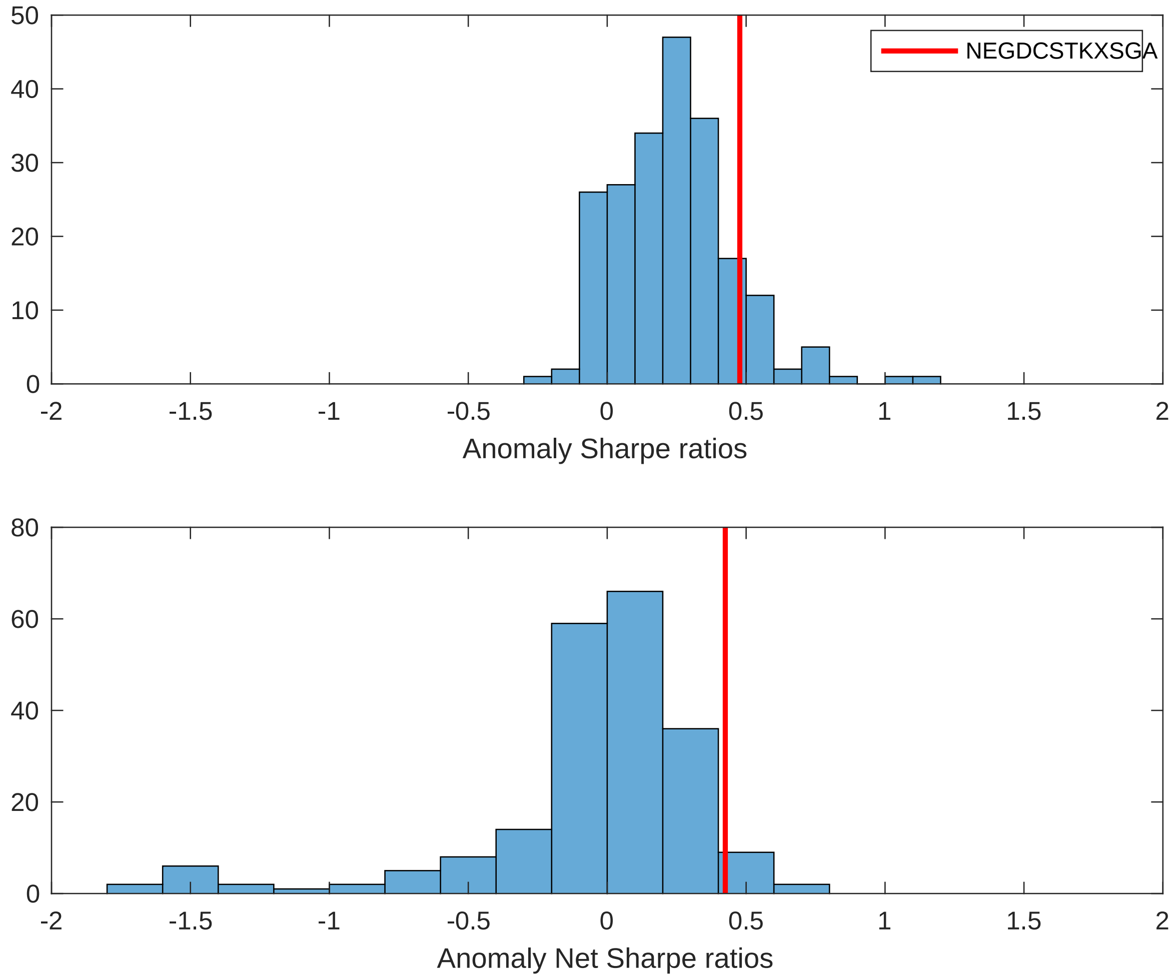


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OEA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

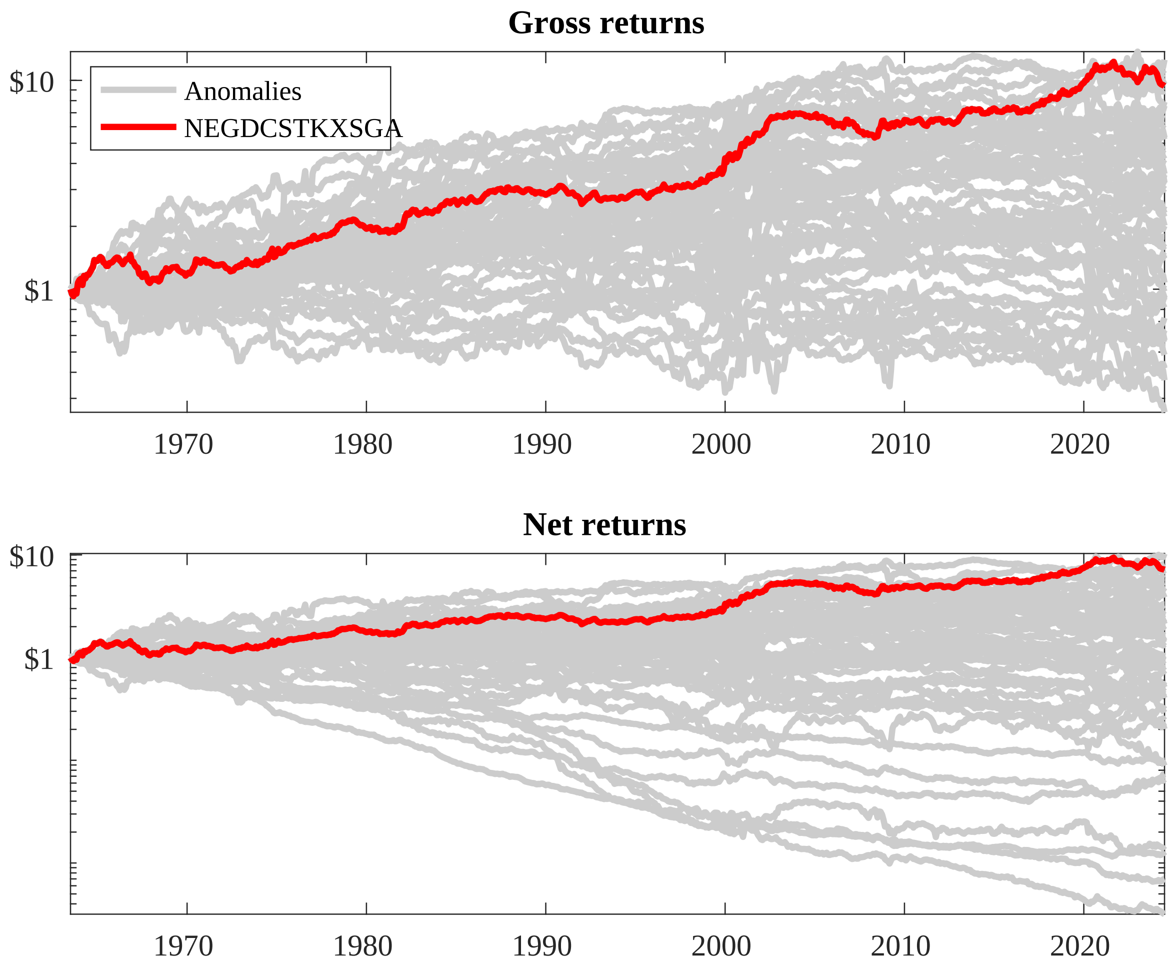
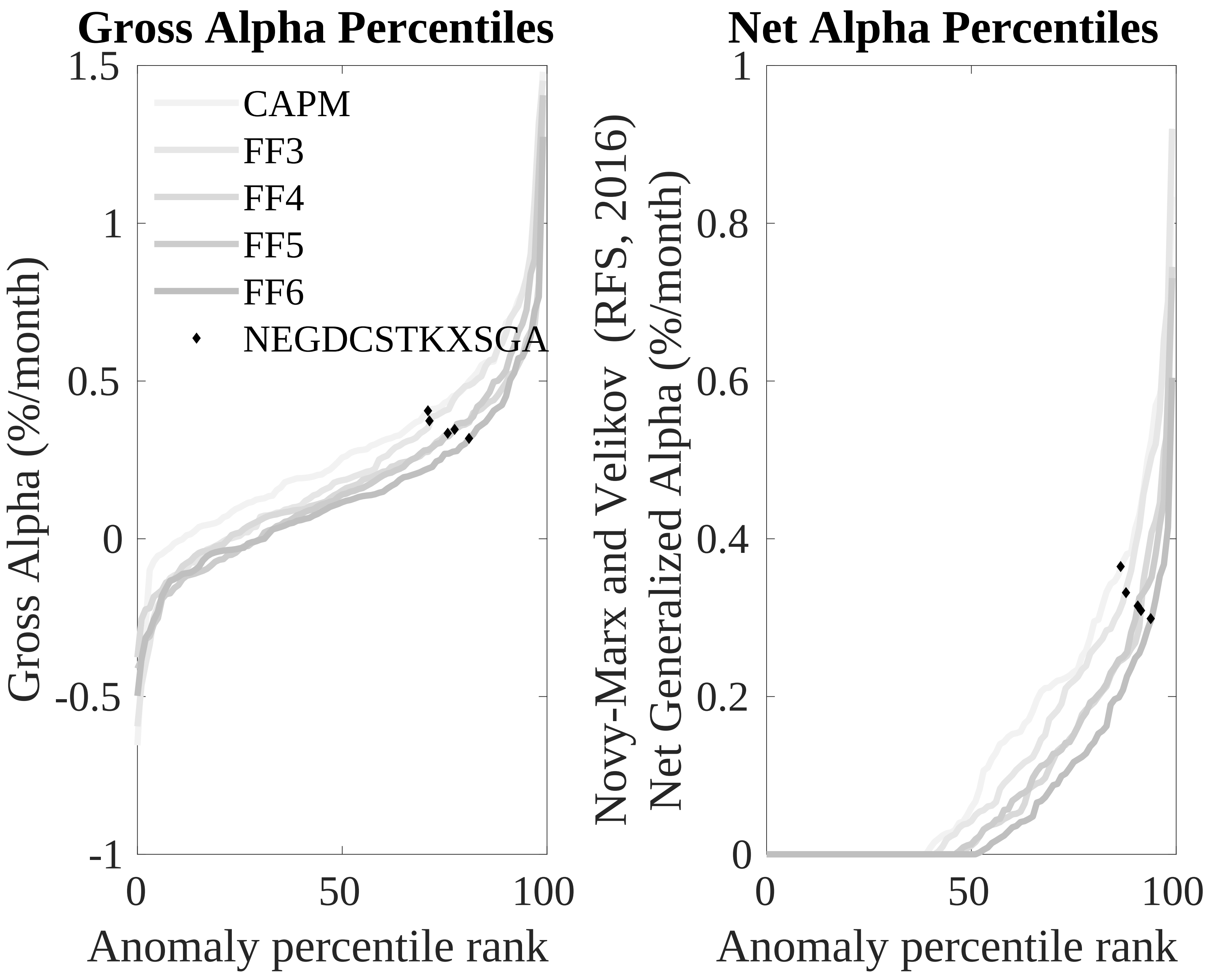


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OEA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKXSGA
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank

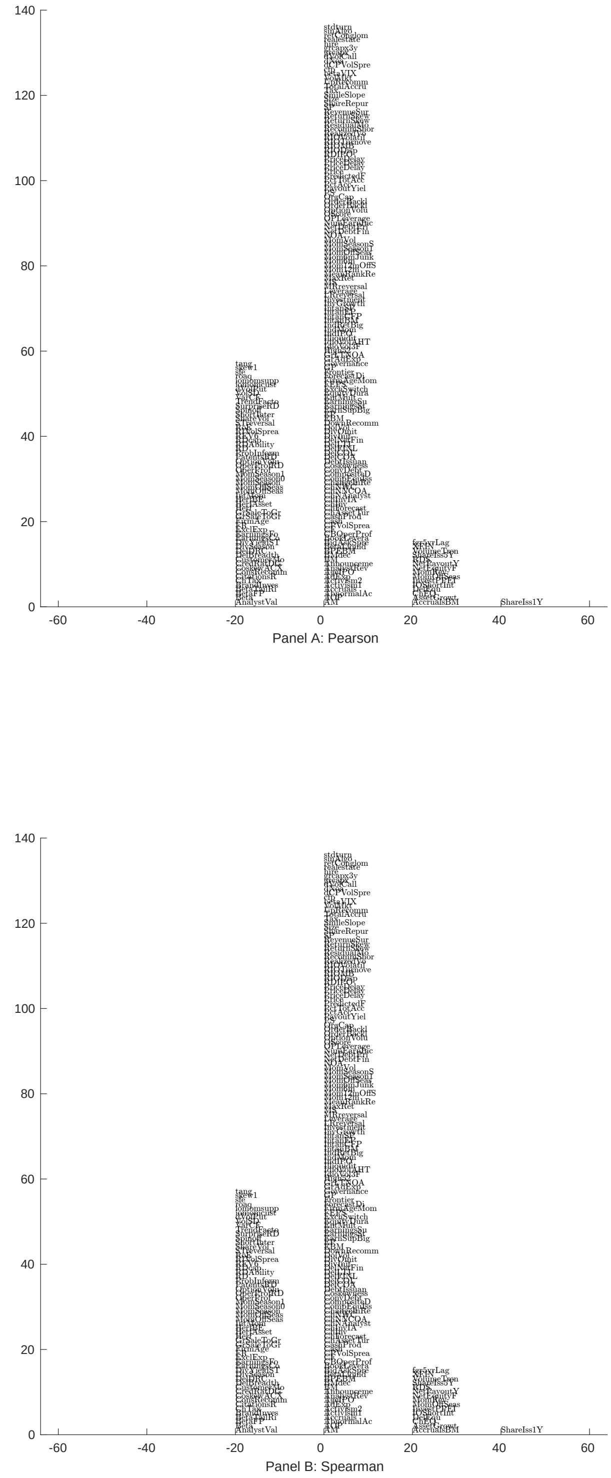


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with OEA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

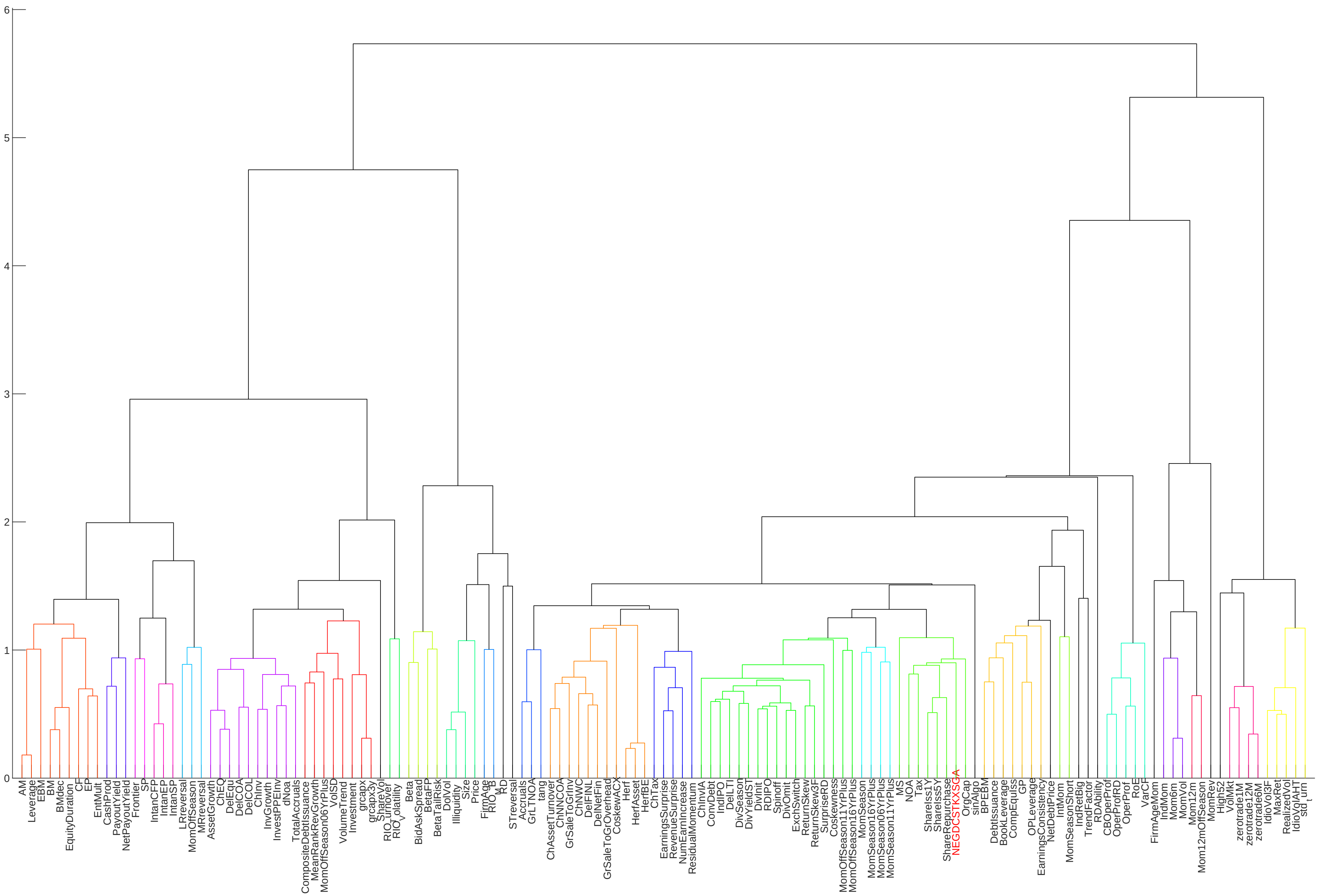


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

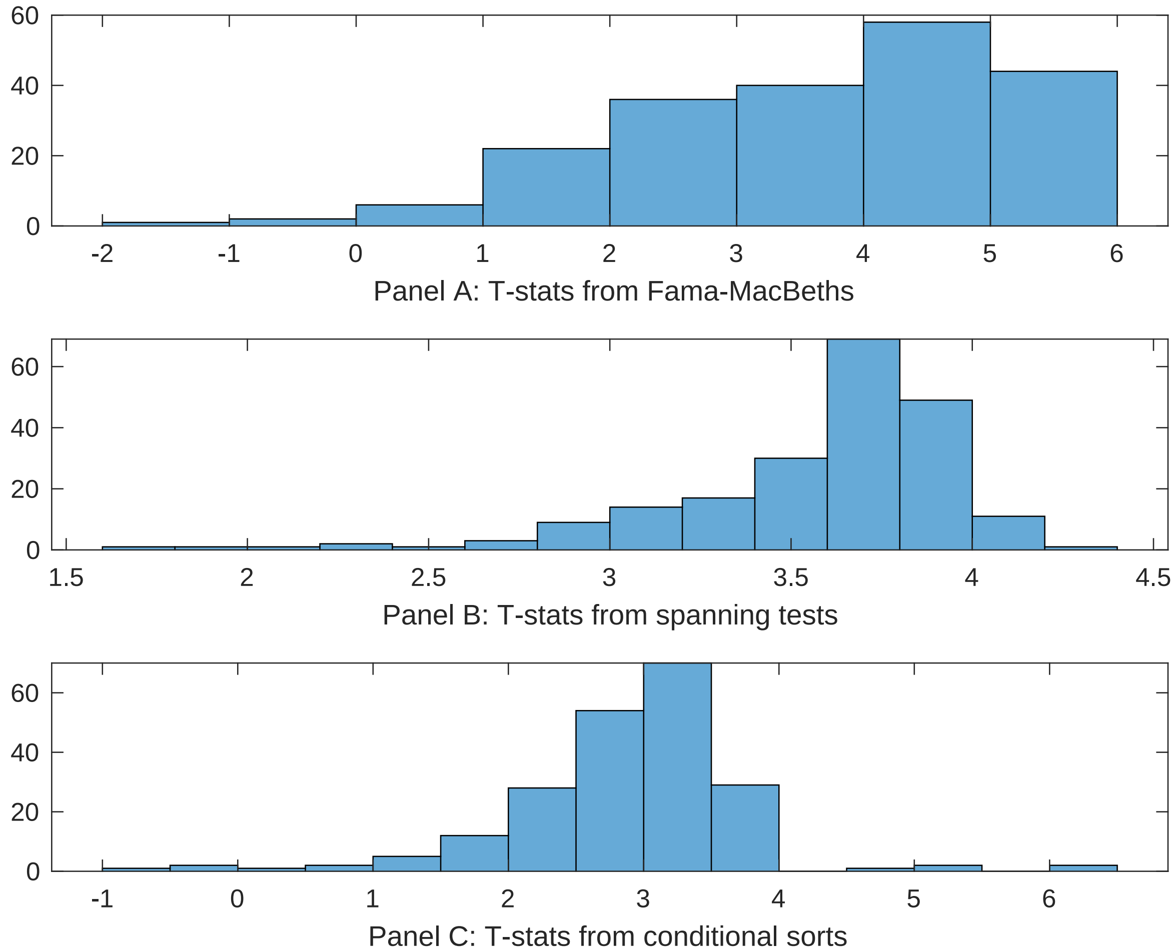


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OEA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OEA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OEA}OEA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.12	0.13	0.13	0.13	0.13	0.12
	[7.22]	[5.26]	[5.89]	[6.26]	[5.60]	[5.16]	[4.49]
OEA	0.79	0.63	0.74	0.62	0.85	0.83	0.41
	[3.48]	[2.21]	[3.55]	[3.01]	[3.72]	[3.02]	[1.45]
Anomaly 1	0.48						-0.16
	[4.55]						[-0.98]
Anomaly 2		0.28					0.15
		[2.67]					[2.91]
Anomaly 3			0.14				0.85
			[5.93]				[3.54]
Anomaly 4				0.27			0.28
				[5.08]			[6.23]
Anomaly 5					0.15		0.13
					[4.34]		[2.07]
Anomaly 6						0.17	-0.14
						[2.64]	[-1.75]
# months	708	703	703	703	708	630	622
$\bar{R}^2$(%)	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the OEA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OEA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.30	0.34	0.30	0.24	0.31	0.34	0.28
	[3.40]	[3.86]	[3.35]	[2.72]	[3.50]	[3.67]	[3.22]
Anomaly 1	34.64						39.13
	[7.27]						[5.59]
Anomaly 2		19.51					13.25
		[5.64]					[3.02]
Anomaly 3			24.08				0.92
			[5.30]				[0.16]
Anomaly 4				27.67			20.71
				[5.80]			[3.64]
Anomaly 5					16.82		-10.37
					[3.76]		[-1.61]
Anomaly 6						12.73	-10.82
						[3.07]	[-2.10]
mkt	-6.59	-4.73	-7.14	-6.50	-8.04	-7.36	-6.71
	[-3.14]	[-2.16]	[-3.34]	[-3.05]	[-3.73]	[-3.26]	[-3.06]
smb	0.51	4.85	2.57	3.49	1.46	9.79	2.92
	[0.17]	[1.54]	[0.83]	[1.13]	[0.47]	[2.73]	[0.83]
hml	-3.81	-7.24	-0.16	2.64	-1.78	0.51	-3.88
	[-0.93]	[-1.66]	[-0.04]	[0.64]	[-0.42]	[0.12]	[-0.93]
rmw	-5.30	-18.39	-20.04	-16.92	-5.39	-9.58	-11.64
	[-1.29]	[-3.97]	[-4.14]	[-3.77]	[-1.28]	[-1.96]	[-2.21]
cma	-9.89	7.39	8.77	10.35	6.75	3.81	-29.81
	[-1.31]	[1.11]	[1.32]	[1.61]	[0.89]	[0.57]	[-3.82]
umd	3.63	5.79	3.14	2.43	4.65	4.18	3.17
	[1.74]	[2.72]	[1.48]	[1.14]	[2.18]	[1.97]	[1.55]
# months	731	727	727	727	731	630	626
$\bar{R}^2$(%)	15	12	12	12	10	8	19

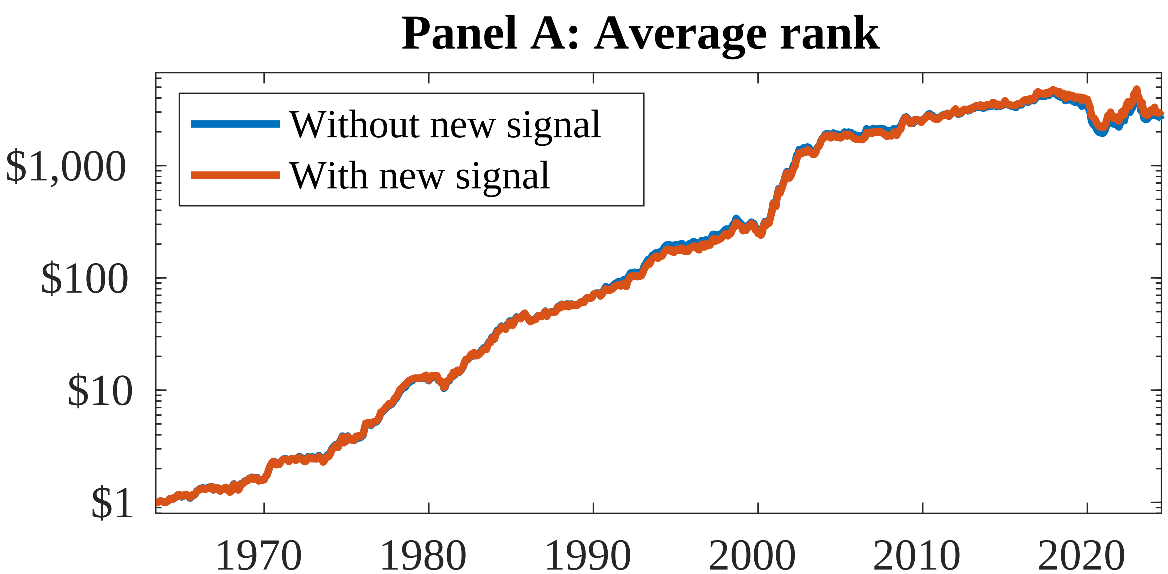


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OEA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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